

on the other hand, has been showing a broader play. For hardware engineers, Vivado is a comprehensive suite of design, analysis and implementation tools.

SDAccel development environment supports OpenCL, C and C++ for software engineers using FPGAs. It enables up to 25 times better performance/watt for data centre application acceleration leveraging FPGAs. High-level synthesis (HLS) provides maximum benefit from the FPGA's hardware programmability.

Altera's OpenCL implementation creates what one might think of as a GPU on an FPGA—with processing elements analogous to GPU cores that execute software kernels in parallel. This solution is easy to program, as it mimics many elements of a GPU, but doesn't achieve the same performance and quality-of-results as Xilinx's HLS-based approach. HLS has been repeatedly shown to deliver hardware architectures on par with what skilled experts can create with hand-coded, manually-optimised RTL.

HLS can quickly generate the RTL code, which synthesises down into your chosen architecture. But encapsulating all that power into something that appears to software engineers as a 'compiler' is a substantial task.

HLS supports several languages (C, C++ and OpenCL), all at the same time, with the same implementation back-end. There are certainly vast application areas that use C and C++ rather than OpenCL. In order to create a 'software-like' development and runtime environment, several pieces of the normal FPGA flow had to be improved.

First, for testing and debugging code, the normal HLS, logic synthesis, place-and-route, timing-closure and device programming cycle would be far too clunky for the rapid-iteration development software engineers are accustomed to. To get around this, Xilinx created

a software environment that would be comfortable to any Eclipse-using programmer, and then set up their HLS tool to spit out software-executable cycle-accurate versions of the HLS architecture. This allows the 'accelerated' algorithm to execute completely in software for debug and testing purposes, without the long logic synthesis and place-and-route runs required for the final hardware-based implementation.

To get around the run-time issues, Xilinx deployed their partial-reconfiguration technology. This allowed the FPGA to be re-configured with a different algorithm without interrupting system execution. This is critical in data centre applications where a server cannot be arbitrarily taken offline in order to process a normal, full-FPGA configuration cycle. Partial reconfiguration allows the FPGA to keep its inputs/outputs live and to keep some baseline communication with the applications processor while it quickly swaps in new acceleration bitstreams. If the software development and deployment environment is good enough to realise even a fraction of that potential in the data centre, it could represent the biggest revolution ever in managing data centre power.

Emerging algorithms are gaining traction in a wide variety of pattern-matching jobs, from serving up Facebook news feeds to piloting self-driving cars. FPGAs are enabling new types of servers to execute massive bitwise operations in parallel. Analytics servers can replace hundreds of contemporary servers. Microsoft's data centres drive services like Cortana and Skype Translator, and the company is constantly looking to upgrade server performance. Microsoft is also working with Intel to implement silicon photonics, which will replace copper wires with fibre optics for faster communication between data centre servers. **EFY**



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